3 Fig. 3.1 shows a pair of identical loudspeakers A and B placed 2.0 m apart and emitting coherent sound waves of frequency 470 Hz. An observer walks from X to Y. The perpendicular distance between the sources and XY is 12 m. As he walks, he hears a sound of maximum intensity at P, followed by minimum intensity at Q and the next maximum intensity at R. P is equidistant from A and B. R is 4.5 m away from P.

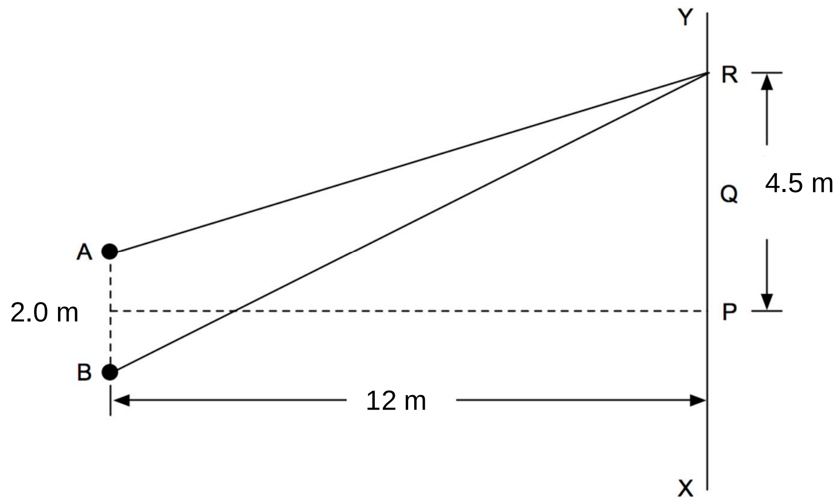

Fig. 3.1

(a) Explain how minimum intensity of sound is formed along the line XY.

....

....

....

.... [2]

(b) (i) Show that BR is 13.2 m.

[1]

(ii) AR is 12.5 m.

Determine the wavelength of the sound.

wavelength = ... m [2]

(iii) Determine the speed of the sound.

speed = ... m s^{-1} [2]

(c) The sound reaching Q from A alone has intensity I and that from B alone has intensity $\frac{I}{3}$.

Determine the intensity at Q in terms of I.

intensity = ... I [3]

4 **(a)** A car headlamp is marked 12 V, 36 W. It is switched on for a 20 minute journey.